

FINAL PROJECT REPORT

Predictive Maintenance for Metro Climate & Air Systems

What 1.5 million sensor readings revealed about how a metro air compressor fails.

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Project: 21 June to 15 August 2027 · Data: Porto Metro APU, Feb to Sep 2020

0.580

final F1

0.985

recall

4 / 4

faults detected

0.97 ms

end-to-end per window

The Project in One Slide

OVERVIEW

Eight weeks, 21 June to 15 August 2027. Two course corrections, both forced by the data.

PROPOSE • 21 JUN

Cycle-level anomaly detection.
Baseline named: F1 0.32.



CHECKPOINT • 18 JUL

Cycle unit failed. Moved to
15-min windows. IF lost to physics.



FINAL • 15 AUG

Two-layer system shipped.
Early warning fell back to CUSUM.

The final system: Layer 1 detects a fault in progress using an Isolation Forest gated by cluster membership. Layer 2 issues an advisory early warning using a CUSUM change-point detector on cycle rate. The two layers exist because they solve genuinely different problems, and only one of them is solved well.

1,516,948

readings processed

16,773

modelling windows

16.25 s

full clean, end to end

2 of 4

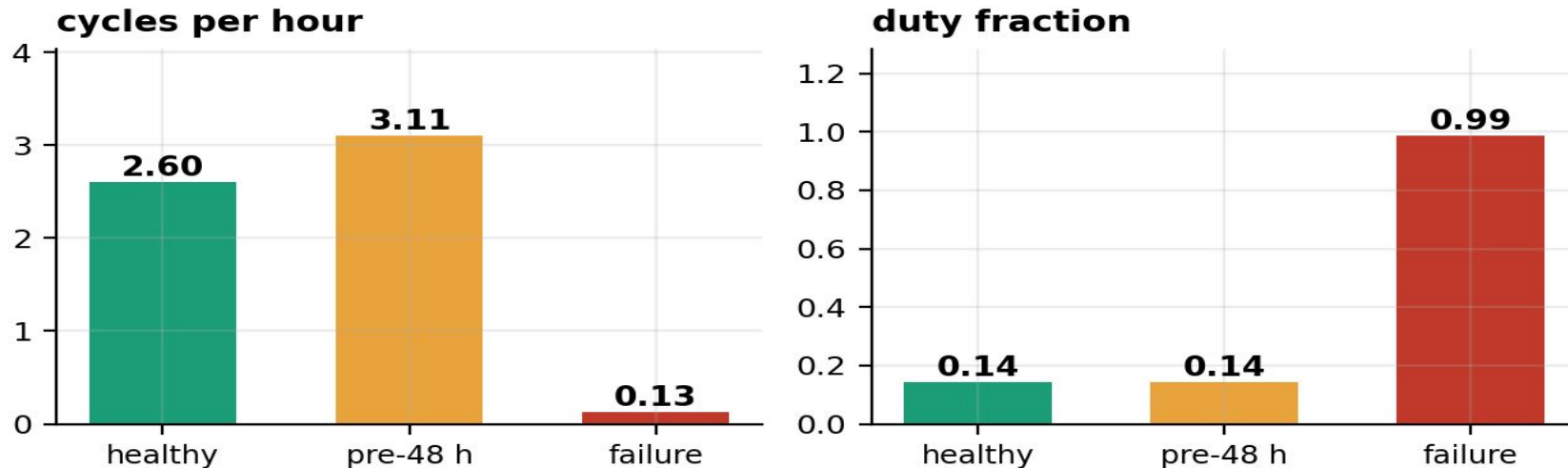
faults warned in advance

Note on the 0.32: it was measured per-sample on the raw stream, not per-window after cleaning. The like-for-like comparator is 0.577, so the real gain in F1 is 0.003, not 0.26. See slide 5.

How This Machine Actually Fails

FINDING 1

A three-stage signature, found in the data, not assumed from theory.



HEALTHY

2.60 cycles/h

long, infrequent refills



DEVELOPING LEAK

3.11 cycles/h (+20%)

reservoir drains faster,
so it refills more often



FAILURE

0.13 cycles/h (-95%)

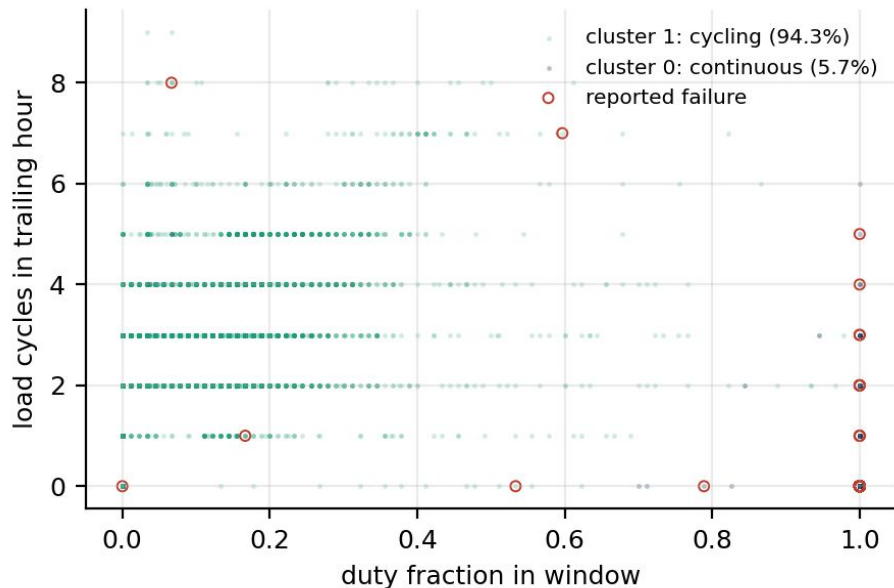
leak outpaces capacity,
cycling stops entirely

Duty cycle sits flat at 0.14 until the moment of failure. Any duty-based alarm therefore has ZERO lead time by construction.

Two Operating Regimes, Found Without Labels

FINDING 2

$k = 2$, silhouette 0.676. The fault regime emerged unsupervised.



Cluster 1 · cycling

94.3% of windows · 2.6 cycles/h, duty 0.11

0.03% are reported faults

Cluster 0 · continuous run

5.7% of windows · 0.004 cycles/h, duty 1.00

34.0% are reported faults

98.5% of faults are in cluster 0

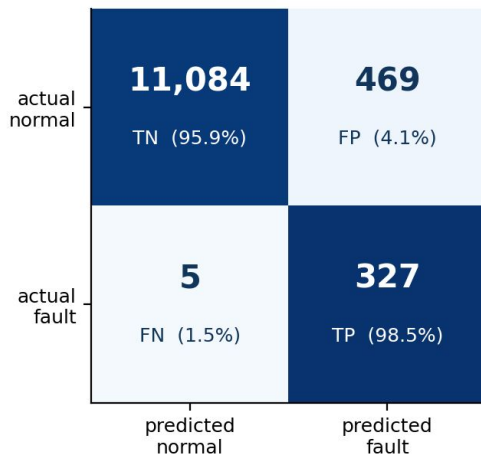
But two thirds of cluster 0 has no maintenance report.
Continuous running is necessary, not sufficient.

This is the ceiling on precision, and it is set by the labels rather than by any model.

Detection Performance

RESULTS

Test period April to September. 11,885 windows, 332 of them faults.



0.580

F1

0.411

precision

0.985

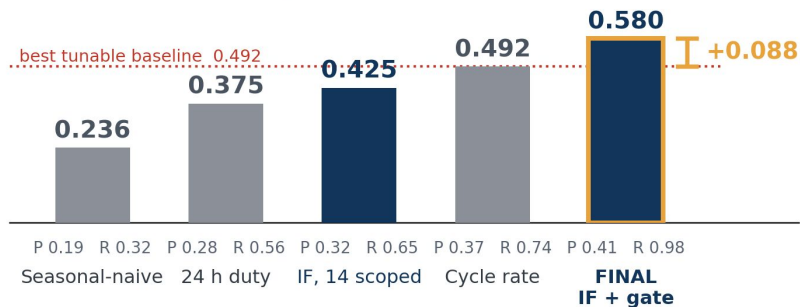
recall

92 / 30d

false alarms

At equal alert budget: 92 false alarms per 30 days

Bar height is F_1 . Instantaneous duty is absent: it saturates on 798 windows and cannot be tuned to a budget.
grey = physical baseline navy = learned model



Why accuracy is the wrong headline

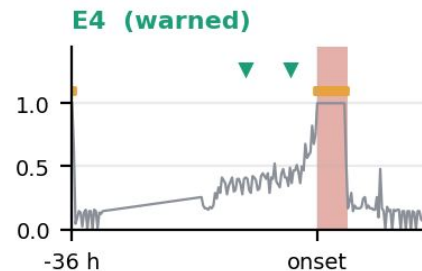
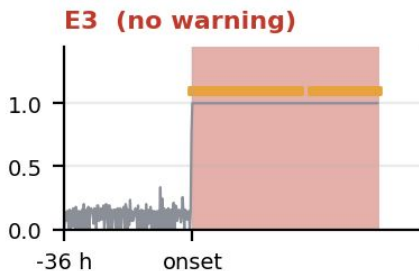
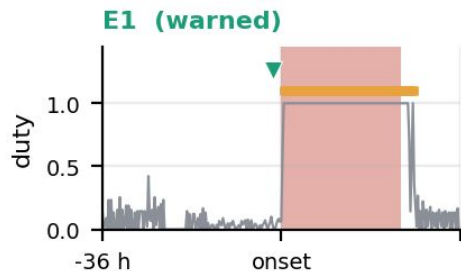
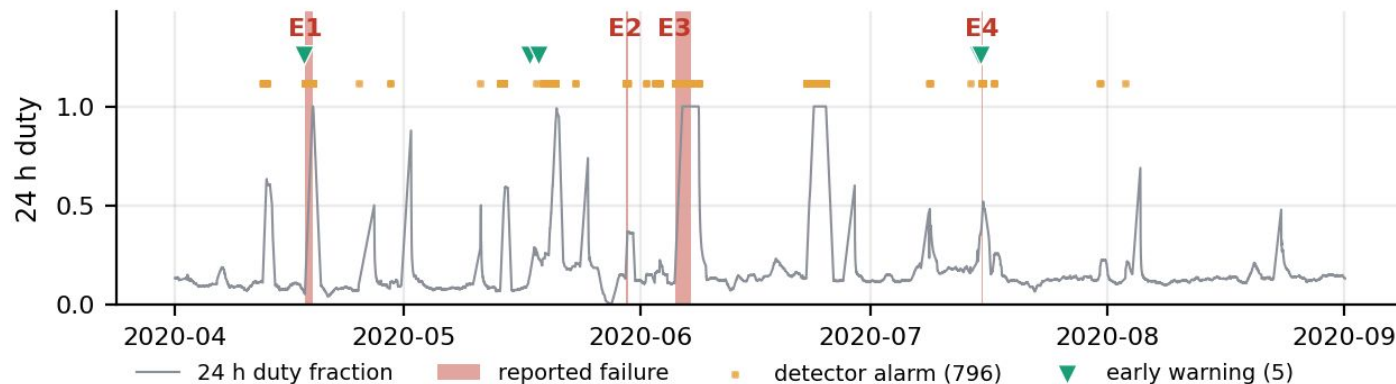
Our model scores 96.0% accuracy. A model that predicts 'normal' every single time scores 97.2%. Accuracy rewards ignoring the faults entirely, which is why every number we report is precision, recall or PR-AUC.

Like for like: the best physical threshold reaches F_1 0.577, so our gain is 0.003, not the 0.26 implied by the proposal's 0.32 baseline. At EQUAL alert budget the margin is 0.088, because duty saturates (798 tied windows) and cannot be dialled to a budget at all. The model buys controllability, not accuracy.

The System Running Across Five Months

EVIDENCE

Every fault caught. Early warning on two of four. Insets zoom three events.



4 / 4

faults detected while in progress

2 / 4

faults warned before onset

10.5 h

longest advance warning

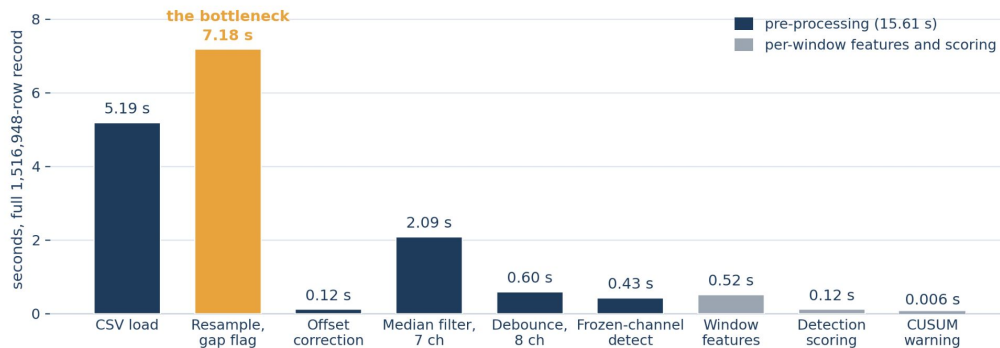
0.4 / 30d

early-warning false alarms

Feasible at Scale

EFFICIENCY

The deployment question is settled, and not narrowly.



Six pre-processing stages sum to 15.61 s. Plus window features and scoring, end to end is 16.25 s, or 0.97 ms per window.

Headroom

0.97 ms

END TO END per window, including all pre-processing, against a 15-minute window period.

About 10^6 times faster than real time. Scoring alone is 0.008 ms, but that is only the cheapest stage.

97,190

readings/s cleaning throughput

15.6 s

to clean all 1.5M readings

7.18 s

resampling and gap flagging, the real bottleneck

2.2 s

18 weekly drift refits, total

We quote the end-to-end figure because it describes the deployed system, and 10^6 already settles the question.

What Worked, What Did Not

Three methods failed. Two of the three taught us something; the third we still cannot fully explain.

Worked

✓ Clustering

0.07 s to fit, recovered the fault regime with no labels. Best value per hour spent in the project.

✓ Physics-scoped features

Cutting 41 features to 14 raised F1 from 0.31 to 0.55. Domain reasoning beat capacity.

✓ Flagging gaps, not filling them

909 hours of missing record never became fabricated training data.

Did not work

✗ Cycle as the modelling unit

Only 3 of 10,974 cycles fall inside a fault, because failure stops the cycling. Demoted to a feature source.

✗ Isolation Forest on all features

F1 0.31, worse than a one-line threshold. Cause: seasonal drift plus feature dilution.

✗ Engineered precursor features

Ten drift-robust features made lead-time prediction slightly worse. At four events we cannot say why.

The switch rule fired at the bar rather than below it: 3.0x vs 2.8x is inside one fold SD, so we resolved the tie conservatively and shipped the fallback. Seven planned analyses were also cut, SHAP attribution first; report section 7.3 lists all of them.

Cluster membership alone matched our F1 and beat our PR-AUC; the head-to-head that would have settled it was in the cut week 7.

Where This Goes Next

NEXT

Four directions, ordered by how much they would change the answer.

1

Instrument cycle timing directly

The strongest early signal is cycle frequency, which we recovered indirectly from a 10 s digital flag. A dedicated cycle-event log at source would sharpen it substantially, and costs almost nothing to add.

2

Get more events

Every supervised limit here is a four-event limit, not a method limit. Pooling MetroPT-1 and MetroPT-2, which cover other vehicles and oil-leak faults, is the highest-value next step.

3

Real-time dashboard

At 0.008 ms per window the scorer is free. A streaming deployment showing regime, anomaly score and cycle-rate CUSUM would close the loop to the maintenance desk.

4

Transfer to other fleets

Test whether cycle-frequency features generalise across compressors and transit systems, or whether each unit needs its own baseline.

Note the ordering. Two of the top three are about better data. Eight weeks established that the models were not the constraint.

What We Learned

About the machine, and about doing this kind of work.

About the compressor

Failure has a three-stage signature. Cycling frequency rises 20% before onset and collapses 95% at onset, while duty cycle stays flat until the fault is already present.

About the modelling

A one-line physical threshold beat a 41-feature Isolation Forest. Domain reasoning outperformed model capacity at every stage of this project.

About the data

Two thirds of the continuous-run regime carries no maintenance report. Precision here is bounded by record-keeping, not by algorithms.

About the process

Two planned methods failed and were replaced. Writing the switch rule down before we knew the answer is what kept week 6 from becoming open-ended.

Where this leaves a maintenance planner: a detector that essentially never misses a fault in progress, and an advisory warning that fires hours ahead on half of them. Half the problem, solved.